

Seat No:

SET	P
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Pune University

Btech Endsem

Subject: Software Testing and Quality Assurance · Pattern: Section 1 24 Questions 70 Marks

Day & Date: Tuesday, September 22, 2026 **Duration:** 180 Minutes **Max. Marks:** 34 Marks

Instructions for Candidates:

1. Q.1 is compulsory. Mention the question paper set (**P**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. 24 Marks Theory Based Questions

Q.1 Choose the correct alternatives for the following questions. [9 Marks]

1. a) Explain why ISO-9001 standard and its importance in software testing? [6]
a) Illustrate Selenium's IDE and explain in detail. [6]
b) Can you clarify different levels of CMM? [5]
2. a) Examine the relationship between quality & productivity. [5] [5]
a) Explain PDCA cycle. [5]
b) Plan software quality control with respect to college attendance software.
3. a) Define software quality. List & explain core components of quality. [5] OR
a) What are the constraints of software product quality assessment? [5]
b) Explain defect life cycle with the help of diagram. [5]
4. a) Illustrate Selenium tool suite in detail. [6]
a) Identify different benefits of Automation testing. [6]
b) How would you explain Selenium WebDriver? Explain it. [6]
5. a) How would you explain Selenium IDE explain in detail? [6] OR
a) Explain Robotic Process Automation in detail. [6]
b) Construct different automated testing processes. [6]
6. a) Write test cases for login validation. [5] documentation. [5] OR
a) What are the entry & exit criteria of testing? [5]
b) Analyse test policy & test strategy which is included in test

7. a) Can you explain path coverage testing & conditional coverage testing. [6]
a) Identify the importance of Regression Testing & explain it. [6]
b) Explain in detail performance testing with example. [6]
8. a) Compare the Ishikawa's Flow chart and Histogram tool. [6] OR
a) Explain in detail six sigma characteristics in details. [6]
b) Can you explain How to maintain SQA? [5]
9. a) What do you think about static techniques? [6] OR
a) State in your own words Error guessing and exploratory testing. [6]
b) How would you explain System Testing and Acceptance Testing? [6]
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SECTION – I (Theory & Core Concepts) [14 Marks]

Q.2 Answer the following questions (Attempt Any Four): [16 Marks]

- a) Explain the Six Sigma methodology and its characteristics in quality management. [6]
b) Explain ISO 9001 and CMM/CMMI quality standards and their relevance to software testing. [6]
c) Explain Statement, Branch, and Path coverage metrics with suitable code examples. [4]
d) Explain the defect life cycle with a neat diagram. [4]
e) Explain the concept of Regression Testing and when it should be performed. [3]

Q.3 Answer the following questions in detail (Attempt Any Two): [12 Marks]

- a) Explain Decision Table based testing and Cause-Effect graphing techniques. [4]
b) Explain Automation Testing frameworks (Data-Driven, Keyword-Driven, and Hybrid) with their benefits and limitations. [6]
c) Explain Equivalence Class Partitioning and Boundary Value Analysis with examples. [4]
d) Differentiate between Black-Box and White-Box testing techniques with examples. [6]
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SECTION – II (Analysis, Design & Applications) [14 Marks]

Q.4 Answer the following questions (Attempt Any Four): [16 Marks]

- a) Compare Error Guessing and Exploratory Testing approaches. [4]
b) Write a short note on Test Plan and Test Strategy documentation. [3]
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- c) Explain Statement, Branch, and Path coverage metrics with suitable code examples. [4]
- d) Explain the defect life cycle with a neat diagram. [4]
- e) Compare Error Guessing and Exploratory Testing approaches. [4]

Q.5 Solve / Explain the following technical problems: [12 Marks]

- a) Explain ISO 9001 and CMM/CMMI quality standards and their relevance to software testing. [6]
- b) Explain the Six Sigma methodology and its characteristics in quality management. [6]
- c) Explain Automation Testing frameworks (Data-Driven, Keyword-Driven, and Hybrid) with their benefits and limitations. [6]

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